

Customer Information

Mr Collins

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Eastern Region, Ghana

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Quote #: QT-COLLINS-E43DC5

Date: May 15, 2026

Valid for: 30 days

Off-Grid Hotel Package

Engineered off-grid package for ~332 kWh/day (after diversity) at Koforidua, Eastern Region. 194× 570W panels, 90.0 kW inverter, 330.0 kWh battery storage (24h essential backup basis).

Item	Unit price (GHS)	Qty	Amount (GHS)
Longi 570W Panel	1,500.00	194.0	291,000.00
PV mid / end clamp (each) (3.5× panels)	30.00	679.0	20,370.00
MC4 connector pair (~1.2 pairs per panel)	25.00	233.0	5,825.00
Energy Precision 10kW 30.0kW Inverter (×3 parallel)	11,000.00	3.0	33,000.00
RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	2,500.00	6.0	15,000.00
125A AC breaker (inverter 30.0 kW > 8 kW)	300.00	1.0	300.00
AC isolator switch 40A (AC isolator per inverter)	280.00	3.0	840.00
100A changeover (1 per inverter / genset path)	1,500.00	3.0	4,500.00
Energy Precisions 16.0kWh Battery	26,000.00	21.0	546,000.00
Battery cable 50mm copper (per meter) (~0.9 m per kWh (large bank))	200.00	302.0	60,400.00
Battery cable 35mm copper (per meter) (supplementary 35 mm runs)	80.00	168.0	13,440.00
250A MCCB battery breaker (battery bank protection)	650.00	34.0	22,100.00
Mounting structure (rails) (proforma rail sets)	350.00	122.0	42,700.00
Transportation of logistics	3,000.00	1.0	3,000.00
Installation & commissioning (large) (tier for 110.6 kW)	8,000.00	1.0	8,000.00
Single core 6mm PV cable (100m coil) (PV homerun (17 DC strings))	2,500.00	9.0	22,500.00

Item	Unit price (GHS)	Qty	Amount (GHS)
AC cable 6 mm² (100m coil) (AC branch runs)	1,800.00	3.0	5,400.00
63A DC MCB 2P (DC string groups (17 strings))	200.00	17.0	3,400.00
DC isolator switch 32A (DC isolator per string group)	350.00	17.0	5,950.00
100A AC breaker (main AC feed)	250.00	1.0	250.00
6-way breaker box	180.00	1.0	180.00
13-way breaker box	350.00	12.0	4,200.00
SPD / lightning arrester (1000V class) (large array)	300.00	1.0	300.00
Earthing kit (rod, clamp, 16mm earth cable) (earthing per array section)	450.00	14.0	6,300.00
PVC trunking 25mm (per metre) (trunking metres)	35.00	291.0	10,185.00
DC/AC labeling & warning kit (labels & warnings)	120.00	1.0	120.00
Miscellaneous (larger kit — Option 2 reference) (remaining consumables (glands, lugs, ties))	2,000.00	1.0	2,000.00

Sub total: 1,127,260.00

Tax (12.5%): 140,907.50

Discount (0%): 0.00

Option total 1,268,167.50 (GHS):

SYSTEM SPECIFICATIONS & ENGINEERING DETAILS

DAILY ENERGY
REQUIREMENT

331.64 kWh/
day

EFFECTIVE DAILY
ENERGY

460.62 kWh/
day

After system losses

MONTHLY
GENERATION

13818.6
kWh

PEAK SUN HOURS

5.0 hrs/day

Location: N/A

PV ARRAY
CAPACITY

110.58 kW

194 × 570W panels

PANEL BRAND

Longi

570W model

INVERTER
CAPACITY

85.0 kW

Selected: 90.0kW
(for quote)

DC/AC RATIO

1.23

Optimized

BATTERY
CAPACITY

330.0 kWh

BACKUP
DURATION

24 hours

REQUIRED ROOF
AREA

605.3 m²

MOUNTING RAILS
(EST.)

90 sticks (18 ft)

≈ 491.4 m linear

SYSTEM
EFFICIENCY

72.0 %

Engineering Notes: This system is designed using industry-standard engineering factors, including system efficiency (accounting for inverter, wiring, and temperature), an optimized DC/AC ratio to prevent inverter clipping, and location-specific peak sun hours for accurate energy-generation calculations. All calculations include appropriate safety margins for reliable performance.

Mounting rail stick count is a **planning estimate** (18 ft segments, portrait grid); confirm with site layout and structural design.

Terms & Conditions

This quotation is valid for 30 days from the date of issue. All prices are subject to change without notice. Installation and project delivery terms will be discussed upon acceptance.

Payment Terms

Commercial off-grid hotel proposal — 30% deposit, 40% on equipment delivery, 30% on commissioning.

Notes

Client: Mr Collins · +233544036297 Site: Eastern Region, Ghana — 40-room hotel, restaurant & kitchen.
Objective: sufficient solar generation to operate off the national grid. Load per client brief: 16 ACs, 38 TVs, 100 bulbs, 36 fridges, 2 freezers, water pump, laundry & kitchen equipment, 55 fans.

Thank you for your business!